

Roger (Shaoze) Li

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EDUCATION

University of California, Santa Cruz, Santa Cruz, CA

Master of Science in Computer Science

Graduate in 2025

Algorithm Analysis, Data Structures, Logical Architecture, Cryptography, Spectral Graph, Network Security, Machine Learning

Bachelor of Science in Computer Science - Dean's Honor

Graduate in June 2024

Computer Systems, Databases, Artificial Intelligence, Computational Models, Computer Graphics, Web Development

PROFESSIONAL EXPERIENCE

5G RAN Software Engineer Intern - AgileRadioCom

Jun 2025 – Present

- Develop and simulate 5G physical layer algorithms using **MATLAB** to analyze performance under various channel conditions.
- Design and simulate high-frequency antenna components for 5G base stations using **Ansys HFSS**.
- Collaborate in an agile startup environment to integrate simulation models with hardware control software.

Generative AI & SRE/DevOps Cloud Intern - G5InfoTech

March 2025 - May 2025

- Engineered efficient **RAG** workflows and integrated **VectorDBs** to enhance semantic search capabilities by **30%**.
- Optimized **LLMs** through meticulous prompt engineering and fine-tuning for specialized project applications.
- Developed and deployed **RESTful APIs** and user interfaces, delivering robust, AI-powered solutions for real-world use cases.

Software Developer Intern - Artificial Intelligence Explainability Accountability (AIEA) Lab

June 2023 - September 2024

- Spearheaded the redesign of an autograder system deployed across multiple CS courses, achieving **50x performance speedup**.
- Developed a comprehensive testing suite that achieved **95% code coverage**, significantly reducing production bugs.
- Optimized model training with **Hugging Face** tools, reducing training time by **20%** and improving resource use.

Software Developer Intern - UCSC Economic Department

January 2024 - June 2024

- Pioneered research project focusing on the impending massification of market algorithms using **Python** and **Otree**.
- Designed and implemented code generation system that accepts human natural language and outputs software via **LLM**.
- Presented product demonstrations to stakeholders, incorporating feedback to drive iterative development.

Teaching Assistant and Computer Science Tutor - UCSC Baskin Engineering

January 2023 - Present

- Tutored groups of students in Foundations of Programming Languages, Computer Architecture, Computer Systems.
- Collaborated with the professor to refine and present course materials, ensuring clarity and alignment with learning objectives.

PERSONAL PROJECTS

Guided ML Pipeline Design using LLM

- Leveraged **GPT-4** and Llama **LLMs** to analyze and synthesize **Kaggle** competition solutions.
- Engineered a system to extract and transform Kaggle dataset descriptions and shared solutions into ML pipeline designs.
- Developed the project using open-source LLMs, particularly Hugging Face, for model training and inference.
- Enhanced pipeline evaluation by comparing with top-ranked code solutions. Utilized **GPT-4** for formatting and analysis, establishing a performance baseline. Achieved a **200% increase** in analysis speed for data scientists.

NoteSwap - Collaborative Note Sharing Platform - [Github](#)

- Developed a Notes Swap **web application** for class notes sharing, enabling users to upload educational content.
- Crafted an intuitive, responsive user interface using **Dart**, ensuring seamless user experience across devices.
- Implemented back-end functionality with **Firebase**, including secure file management and user authentication.

KEY SKILLS

Programming Languages: Python, JavaScript, C, C++, Java, Dart

AI & Machine Learning: PyTorch, Hugging Face, NLP, RAG, VectorDBs

Cloud & DevOps: AWS (EC2, S3, RDS, Lambda, IAM, CloudWatch), Git, Linux, CI/CD, Docker

Frameworks & Libraries: React, Flask, Node.js, Firebase, oTree, Pandas, NumPy, BeautifulSoup

Tools & Methodologies: MATLAB, Ansys HFSS, Jira, Agile/Scrum Methodologies